

Scenario 17

An online gaming company wants to ensure that game traffic for its premium users is routed through the lowest-latency path in their network, while standard user traffic can take a slightly longer but less congested path.

Question: How would a proactive SDN controller implement this policy? Describe the type of flow rule it would install.

Answer:

A proactive SDN controller installs flow rules into network switches prior to packet arrival, avoiding control-plane delays for new flows. To enforce differential traffic routing based on user tiers, the controller performs the following actions:

- **1. Proactive Path Computation:**

The controller calculates two distinct forwarding paths across the network topology in advance:

- **Low-Latency Path:** The shortest, lowest-delay route reserved for premium users.
- **Standard Path:** An alternative, less congested route designed for standard users.

- **2. Flow Rule Specifications:**

The controller proactively generates and installs specific flow rules in the switches' flow tables along both paths. Each flow rule consists of:

- **Match Fields:** Differentiates user traffic using headers such as Source/Destination IP subnets, VLAN IDs, DSCP (DiffServ) markings, or specific L4 port numbers associated with premium tier users versus standard users.
- **Priority Field:** Premium flow rules are assigned a higher numerical priority value than standard rules to ensure matching precedence.
- **Action Field:** Instructs the switch on forwarding behavior:
 - *For Premium Traffic:* Forward out the specific egress port corresponding to the lowest-latency path (e.g., Output: Port 1).
 - *For Standard Traffic:* Forward out the egress port corresponding to the alternative path (e.g., Output: Port 2).

- **3. Data Plane Execution:**

When packets arrive at the ingress switch, they are matched immediately in hardware against the pre-installed flow rules. Premium packets are instantly directed along the lowest-latency path with zero controller interaction time, ensuring minimal latency and zero jitter for premium gaming sessions.